CCEE Advanced Analytics Using Statistics

Sessions 13 & 14: Statistical Inference & Hypothesis Testing

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1. CORE TERMINOLOGY

Null Hypothesis (H_0)

Default assumption claiming **no effect, no difference exists.** We try to reject it; never "accept" it.

- Example: "New drug has NO effect" or "Two groups have EQUAL means"

Alternative Hypothesis (H_1 or H_a)

Competing claim asserting an effect or difference **exists**.

- **Two-tailed (non-directional)**: H_1: $\mu \neq \mu_0$ → "Different" (standard, safer)
- **One-tailed (directional)**: H_1: $\mu > \mu_0$ or $\mu < \mu_0$ → "Greater/Less" (more specific power)

Significance Level (α)

Maximum allowed probability of Type I error; threshold for rejecting H_0.

- Common: $\alpha = 0.05$ (5%), 0.01 (1%), 0.10 (10%)
- **95% CI ↔ $\alpha = 0.05$ (total); two-tailed splits to $\alpha/2 = 0.025$ per tail**

Confidence Interval (CI)

Range of plausible values for population parameter.

- **Correct interpretation**: "This method captures true parameter 95% of the time in repeated sampling"
- **Wrong interpretation**: "95% chance true value is in THIS interval"
- Formula: Point Estimate ± (Critical Value × Standard Error)
- **Relationship**: 95% CI ↔ $\alpha = 0.05$; 99% CI ↔ $\alpha = 0.01$; Confidence + Significance = 100%

P-Value

Probability of observing data **as extreme as obtained, GIVEN H_0 is true**.

- NOT the probability H_0 is true
- Smaller p-value = Stronger evidence against H_0
- **Decision rule**: If p-value ≤ α → Reject H_0 (significant); If p-value > α → Fail to reject H_0

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2. TYPE I & TYPE II ERRORS

Error Type	Definition	Consequence	Probability
Type I (False Positive)	Reject TRUE H_0	False alarm; conclude effect when none exists	α
Type II (False Negative)	Fail to reject FALSE H_0	Miss real effect; conclude no effect when it exists	β

Power of test = 1 - β = Probability of correctly rejecting FALSE H_0 (detecting real effect)

- Factors: Larger sample size, larger effect size, higher α → Higher power
- Target: Power ≥ 0.80 (80% chance to detect effect if it exists)

Real-world analogy (Medical Test):

- Type I: Healthy person tests positive (false alarm)
- Type II: Sick person tests negative (miss disease—dangerous)

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3. TAILS OF TEST

One-Tailed (Directional)

- H_1 specifies direction: $\mu > \mu_0$ or $\mu < \mu_0$
- All α concentrated on ONE side
- **Advantage**: More power to detect effect in that direction
- **Disadvantage**: Cannot detect effect in opposite direction
- Example: "Does training INCREASE scores?" (directional)

Two-Tailed (Non-Directional)

- H_1: $\mu \neq \mu_0$ (no direction specified)
- α split: $\alpha/2$ on each tail
- **Advantage**: Detects effects either direction (safer, standard)
- **Disadvantage**: Less power than one-tailed
- Example: "Do male/female students have DIFFERENT scores?" (non-directional)

Decision Rule Example: p-value = 0.03, α = 0.05 → Since 0.03 < 0.05 → **Reject H_0** (significant)

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4. PARAMETRIC TESTS: t-TEST

Parametric tests assume data follow some distribution (usually normal) and compare **means**.

One-Sample t-Test

Compare sample mean vs known population mean.

- Example: Is average height = 5.8 ft?
- Formula: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
- Assumptions: Normal data, independence, continuous scale

Independent Two-Sample t-Test

Compare means of **two independent groups**.

- Example: Male vs female average test scores
- Assumptions: Both groups normal, equal variances, independence, continuous
- Check equal variances first: Levene's test ($p > 0.05$ → use standard t; $p \leq 0.05$ → use Welch's t)

Paired t-Test

Compare same subjects measured **twice** (before/after, matched pairs).

- Example: Do students improve after tutoring (same students, before and after)?
- Formula: $t = \frac{\bar{d}}{s_d/\sqrt{n}}$ where d = differences
- **Why**: Eliminates individual differences → More power

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5. PARAMETRIC TESTS: ANOVA

One-Way ANOVA compares means of **3 or more** independent groups.

Logic

Total Variation = Between-Group Variation + Within-Group Variation

- **Between-group**: Differences in group means
- **Within-group**: Spread of data inside each group
- **F-statistic**: $F = \frac{MS_{between}}{MS_{within}}$
- Large F → Groups differ more than expected by chance → Small p-value

When Used

Compare average marks across 3 different coaching institutes, or comparing 4 fertilizer treatments on crop yield.

Key Points

- Hypotheses: H_0: $\mu_1 = \mu_2 = \mu_3$ (all equal); H_1: At least one differs
- Assumptions: Normality, equal variances, independence
- **Important**: ANOVA tells IF difference exists (yes/no), NOT WHICH groups differ
- **Post-hoc tests** (Tukey, Bonferroni) needed to find which groups differ

Avoiding Multiple Testing Problem

Running 3 separate t-tests (comparing 3 groups pairwise) inflates Type I error: $1 - (0.95)^3 \approx 0.14$ (14% error!). Use one ANOVA instead to keep error at $\alpha = 0.05$.

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6. NON-PARAMETRIC TESTS: CHI-SQUARE

Non-parametric tests do NOT assume normality; work with **categorical data** (counts/frequencies).

Chi-Square Test of Independence

Tests **association between two categorical variables**.

- Example: Gender (M/F) associated with product preference (A/B)?
- Data format: Contingency table (cross-tabulation)
- H_0: Variables independent; H_1: Variables associated

Formula

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where O = Observed count, E = Expected count (if independent)

Calculate Expected Counts

$$E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

Assumptions

- Expected count ≥ 5 in each cell (if violated: combine categories or use Fisher's exact test)
- Independence of observations
- Categorical data

Decision

- If p-value ≤ α → Reject H_0 → Variables associated
- If p-value > α → Fail to reject H_0 → No association

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7. NON-PARAMETRIC TESTS: MANN-WHITNEY U

Compares **two independent groups** when data is **ordinal or non-normal**.

When Used

- Satisfaction ratings (1-5 scale) differ between App A and App B?
- Test scores non-normally distributed
- Small sample size with skewed data
- **Alternative to**: Independent t-test when assumptions violated

Key Idea: Rank-Based

1. Combine data from both groups
2. Rank all observations (smallest = 1, largest = n)
3. Sum ranks for each group
4. Compute U statistic
5. If one group ranks systematically higher → U extreme → Small p-value → Reject H_0

Hypotheses

- H_0: Two groups have same distribution (medians approximately equal)
- H_1: Groups have different distributions

Advantages Over t-Test

- No normality assumption
- Robust to outliers
- Works with ordinal data
- Smaller sample sizes acceptable

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8. HYPOTHESIS TESTING WORKFLOW (7 Steps)

1. **Formulate Hypotheses** → State H_0 (no effect) and H_1 (effect exists)
2. **Set α & Choose Tails** → Usually $\alpha = 0.05$; decide one-tailed or two-tailed
3. **Choose Test** → Based on: groups (1/2/3+), data type (continuous/categorical), normality, independence
4. **Calculate Test Statistic** → Compute t, F, χ^2, U, or other from sample data
5. **Find P-Value** → Compare statistic to reference distribution
6. **Make Decision** → If $p \leq \alpha$, reject H_0; if $p > \alpha$, fail to reject H_0
7. **Interpret in Context** → State conclusion in research terms; note limitations

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9. TEST SELECTION DECISION TREE

How many groups are you comparing?

ONE GROUP
→ Comparing to known value? → **One-Sample t-Test** (continuous, normal)

TWO GROUPS
→ Continuous, both normal? → **Independent t-Test** (if independent groups)
→ Continuous, both normal? → **Paired t-Test** (if same subjects, before/after)
→ Continuous, non-normal/ordinal? → **Mann-Whitney U-Test**
→ Categorical data? → **Chi-Square Test**

THREE+ GROUPS
→ Continuous, all normal? → **One-Way ANOVA**
→ Continuous, non-normal? → **Kruskal-Wallis Test** (non-parametric alternative)
→ Categorical data? → **Chi-Square Test**

Quick Checklist Before Choosing:

- How many groups? (1, 2, 3+)
- What scale? (Continuous, Ordinal, Categorical)
- Normal distribution? (Check histogram, Q-Q plot, Shapiro-Wilk test)
- Equal variances? (Levene's test)
- Independent observations? (Study design question)
- Outliers present? (May suggest non-parametric)

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10. TEST COMPARISON TABLE

Feature	t-Test	ANOVA	Chi-Square	Mann-Whitney U
Groups	2	3+	2+	2
Data Type	Continuous	Continuous	Categorical	Ordinal/Continuous
Assumption	Normal	Normal	None required	None required
Basis	Means	Means	Frequencies	Ranks
Parametric?	Yes	Yes	No	No
When to Use	2 normal groups	3+ normal groups	Association testing	Non-normal or ordinal

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11. FORMULAS AT A GLANCE

One-Sample t: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$, df = n - 1

Two-Sample t: $t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}}$, df = $n_1 + n_2 - 2$

Paired t: $t = \frac{\bar{d}}{s_d/\sqrt{n}}$, df = n - 1

ANOVA F: $F = \frac{MS_{between}}{MS_{within}}$, df = (k-1, n-k)

Chi-Square: $\chi^2 = \sum \frac{(O-E)^2}{E}$, df = (rows-1)(columns-1)

CI (General): Point Estimate ± (Critical Value × Standard Error)

Relationships:

- Power = 1 - β
- Confidence Level + Significance Level = 100%
- 95% CI ↔ α = 0.05; 99% CI ↔ α = 0.01

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12. COMMON MCQ TRAPS

Trap 1: Type I vs II

- ✗ Type I = Failing to reject false H_0
- ✓ Type I = Rejecting TRUE H_0 (false positive); Type II = Failing to reject FALSE H_0 (false negative)

Trap 2: One-Tailed vs Two-Tailed

- ✗ Always assume two-tailed
- ✓ "Different/change" → Two-tailed; "Greater/Less/Better" → One-tailed

Trap 3: p-Value Interpretation

- ✗ p-value = Probability H_0 is true
- ✓ p-value = P(data this extreme | H_0 true)

Trap 4: ANOVA Results

- ✗ "ANOVA tells which groups differ"
- ✓ ANOVA tells IF difference exists; use post-hoc tests to find WHERE

Trap 5: Test Selection

- ✗ "I need to compare groups" → Use t-test
- ✓ Ask systematically: How many groups? What data type? Normal?

Trap 6: CI and Alpha

- ✗ "95% CI means α = 0.95"
- ✓ 95% CI ↔ α = 0.05 (inverse relationship)

Trap 7: Chi-Square Expected Counts

- ✗ "All expected counts must be ≥ 10"
- ✓ Expected ≥ 5 acceptable; if violated, combine categories

Trap 8: Power and Sample Size

- ✗ "Small p-value always means large effect"
- ✓ Larger n → Smaller p-values for same effect; use effect size for practical significance

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13. KEY TAKEAWAYS FOR CCEE

1. **Type I error** = Reject TRUE H_0 = Probability α (false positive)
2. **Type II error** = Fail to reject FALSE H_0 = Probability β (false negative)
3. **Power** = 1 - β = Ability to detect true effect
4. **Parametric** (t, ANOVA): Assume normality, mean-based, more powerful
5. **Non-Parametric** (Chi-Square, U): No normality, rank/count-based, robust
6. **t-Test** = 2 independent/paired groups; **ANOVA** = 3+ groups; **Chi-Square** = Categorical association; **Mann-Whitney U** = 2 groups, non-normal/ordinal
7. **ANOVA** tests IF difference exists (post-hoc finds WHERE)
8. **p-value** ≤ **α** → Reject H_0 → Significant; **p-value** > **α** → Fail to reject H_0 → Not significant
9. **95% CI** ↔ **α = 0.05**; **99% CI** ↔ **α = 0.01**; **Confidence + Significance = 100%**
10. **One-tailed** = More power one direction; **Two-tailed** = Detects either direction (standard)
11. **Effect size** ≠ **Statistical significance** (large effect possible even with p > 0.05 if small n)
12. **Post-hoc tests** (Tukey, Bonferroni) run AFTER significant ANOVA to identify which groups differ
13. **Expected counts** ≥ **5** for Chi-Square; if violated, combine categories
14. **Paired t-test** for same subjects measured twice; **Independent t-test** for different subjects
15. Multiple t-tests inflate Type I error; use ANOVA to keep error at α level

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14. REAL-WORLD ANALOGIES

Hypothesis Testing as Court Trial

Concept	Court	Statistics
Default	"Innocent until proven guilty"	H_0 (no effect)
Burden of proof	Prosecution	H_1 (alternative)
Evidence strength	Testimony, physical evidence	p-value
Judge's threshold	"Beyond reasonable doubt"	α (significance level)
Wrong verdict 1	Convicting innocent	Type I error
Wrong verdict 2	Acquitting guilty	Type II error

Decision: If evidence strong ($p \leq \alpha$), convict (reject H_0); else acquit (fail to reject H_0)

Why ANOVA vs Multiple t-Tests?

Running 3 separate t-tests comparing 3 groups pairwise:

- Each test: $\alpha = 0.05$ (5% error)
- Combined: $1 - (0.95)^3 \approx 0.14$ (14% error—too high!)
- **Solution**: One ANOVA keeps overall error at $\alpha = 0.05$

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15. PRACTICE MCQS (Solutions Included)

Q1: Type I error represents:
A. Failing to reject false H_0 B. Rejecting true H_0 C. Accepting H_0 D. Sample mean = population mean
Answer: B

Q2: For 95% confidence interval, α in two-tailed test is:
A. 0.01 B. 0.05 C. 0.10 D. 0.025
Answer: B (though 0.025 is $\alpha/2$ on each tail; 0.05 is total)

Q3: Comparing 3 coaching institutes' average marks. Best test?
A. Three t-tests B. One-Way ANOVA C. Chi-Square D. Mann-Whitney U
Answer: B

Q4: Satisfaction ratings (1-5 scale), non-normal, two stores. Best test?
A. Independent t-test B. Paired t-test C. Mann-Whitney U D. Chi-Square
Answer: C

Q5: p-value = 0.03, $\alpha = 0.05$, two-tailed. Decision?
A. Reject H_0 B. Fail to reject H_0 C. Increase α D. Inconclusive
Answer: A

Q6: Chi-Square tests:
A. Means of groups B. Association between categorical variables C. Medians D. Variances
Answer: B

Q7: ANOVA F = 1.5, p = 0.20, $\alpha = 0.05$. Conclusion?
A. All means equal B. At least one mean differs C. Insufficient evidence to conclude means differ D. Type I error
Answer: C

Q8: Paired t-test used when:
A. Two independent groups B. Same subjects measured twice C. Categorical data D. Expected < 5
Answer: B

Q9: Expected count < 5 in Chi-Square. Action?
A. Use Chi-Square anyway B. Combine categories C. Use Fisher's exact test D. B or C
Answer: D

Q10: Power = 0.85, β equals:
A. 0.85 B. 0.15 C. 0.05 D. 0.50
Answer: B

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16. EXAM STRATEGY QUICK TIPS

Before Test:

- ✓ Memorize Type I/II, p-value, α definitions
- ✓ Practice test selection decision tree
- ✓ Know when parametric vs non-parametric
- ✓ Memorize degrees of freedom formulas

During Exam:

1. **Read carefully** → Identify: groups? data type? assumptions?
2. **Use elimination** → Eliminate obviously wrong answers
3. **Watch for traps** → Type I/II, one-tailed/two-tailed confusion
4. **Remember**: $p \leq \alpha \rightarrow$ Reject H_0
5. **Manage time** → ~1.5 min per question; mark difficult ones to return

Test Selection Anchor:

- 3+ groups + continuous + normal → ANOVA (NOT t-test)
- Categorical data → Chi-Square
- Ordinal/non-normal → Non-parametric
- Same subjects twice → Paired t-test

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